
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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Anonymous submission

ABSTRACT

Multimodal Large Language Models (MLLMs) face significant computational overhead when processing long videos due to the massive number of visual tokens required. To improve efficiency, existing methods primarily reduce redundancy by pruning or merging tokens based on importance or similarity. However, these approaches largely overlook a critical dimension of video content, i.e., changes and turning points, and they lack a collaborative model for spatio-temporal relationships. To address this, we propose a new perspective: similarity is for identifying redundancy, while difference is for capturing key events. Based on this, we designed a training-free framework named ST-SimDiff. We first construct a spatio-temporal graph from the visual tokens to uniformly model their complex associations. Subsequently, we employ a parallel dual-selection strategy: 1) similarity-based selection uses community detection to retain representative tokens, compressing static information; 2) temporal difference-based selection precisely locates content-changing points to preserve tokens that capture key dynamic shifts. This allows it to preserve both static and dynamic content with a minimal number of tokens. Across three long-video benchmarks (VideoMME, LongVideoBench, and EgoSchema) and two distinct base models (LLaVA-Video-7B and NVILA-Video-8B), ST-SimDiff ranks first on essentially every setting we test: retaining only 30% of the tokens, it already reaches 63.2% on VideoMME with LLaVA-Video-7B, matching the 63.3% full-token upper bound while using less than a third of the tokens, and at a 50% budget it reaches 63.8%, actually surpassing the full-token model. The margin over the strongest prior method is consistent rather than incidental: for example, at 30% tokens ST-SimDiff improves over the best competing baseline by +1.9 points on VideoMME and +1.5 points on LongVideoBench. These accuracy gains come with clear efficiency benefits, cutting inference time by up to 30.2% and peak GPU memory by 31.7% on 128-frame videos relative to the uncompressed model. Our code will be released upon acceptance.

1 INTRODUCTION

The rise of Large Language Models (LLMs) has significantly propelled advancements of Large Vision-Language Models (LVLMs), which have demonstrated remarkable capabilities in image and video understanding (Zhang et al., 2024c; Liu et al., 2024c). For video processing, current LVLMs typically sample a video into a sequence of frames and then convert each frame into hundreds or even thousands of visual tokens for the LLM (Bai et al., 2025; Li et al., 2024). While this paradigm is effective, the number of tokens grows explosively with increasing video duration and resolution, and this growth directly drives the cost of inference: as we quantify later, the uncompressed LLaVA-Video model’s inference time rises from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames, with peak GPU memory climbing from 20.9 GB to 35.0 GB over the same range. Such prohibitive computational and storage burdens severely limit the application of LVLMs in scenarios such as long-video analysis and real-time interaction (Fu et al., 2024b).

To address this challenge, various methods are proposed to enhance the efficiency of LVLMs. These approaches fall into two families that also serve as the two baseline groups against which we later

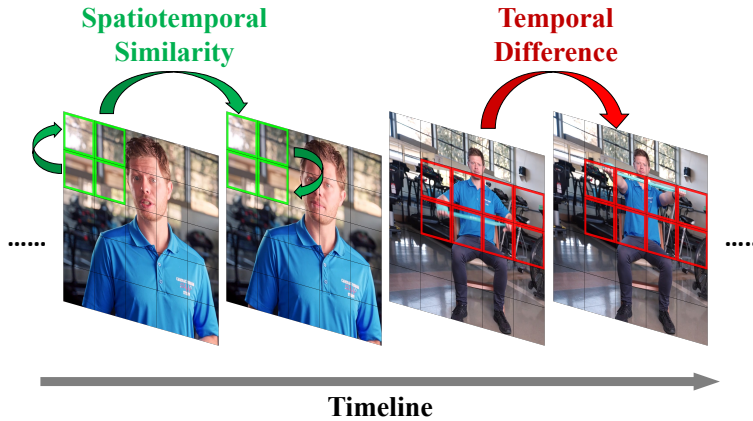


Figure 1: The core motivation of ST-SimDiff. We posit that efficient video understanding requires handling two scenarios simultaneously: Spatiotemporal Similarity (left) can be utilized to compress redundant information both spatially within a frame and temporally across adjacent frames. Temporal Difference (right) should be detected to capture the key actions or events that define the plot.

benchmark ST-SimDiff. The first is importance-based pruning. As observed by FastV (Chen et al., 2024), there is redundancy in the attention distribution across different layers when LVLMs process visual information. Therefore, visual tokens with lower attention scores in the deeper layers of the model can be pruned to reduce the computational load. The second is similarity-based merging/selection, which exploits the high similarity among visual tokens either between adjacent video frames or within a single frame: FrameFusion (Fu et al., 2024b) reduces redundancy by merging similar tokens from adjacent frames, while VisionZip (Yang et al., 2025) directly selects dominant tokens at the vision encoder level. As our experiments show, the strongest of these prior methods (e.g., FrameFusion) remains a demanding point of comparison, yet ST-SimDiff still improves over it by margins that hold consistently across both budgets and both base models.

Despite the success of existing methods in visual token reduction, they share two limitations that our comparisons make measurable. First, existing token pruning methods often focus on spatial correlations within the same frame or temporal correlations at the same positions across different frames. This lack of joint modeling and analysis of spatio-temporal correlations prevents them from effectively capturing complex dynamic events. Second, these methods share a potential blind spot: they mainly focus on informational commonalities, like similarity and importance, while neglecting crucial changes and differences. Since the narrative of the video is often driven by turning events, a model that only seeks similarity might smooth over a sudden action, leading to a misinterpretation of the content. Both limitations surface directly in the numbers: our ablation isolates the joint spatio-temporal modeling and the difference module and shows that each contributes a distinct, additive gain, together lifting VideoMME accuracy from a 60.3% importance-only baseline to 63.2% at a 30% token budget.

Based on this, we propose a new perspective: similarity is for identifying redundancy, while difference is for capturing key moments. We believe that an ideal video token compression algorithm should achieve two goals simultaneously: representing the stable content of a video with the fewest tokens, and precisely preserving the key changes. To this end, we propose **ST-SimDiff**, a dual token selection framework based on a spatio-temporal graph. Specifically, the framework consists of two parts. The first is **similarity-based representative token selection**: we treat all visual tokens of a video as nodes and construct a graph based on their feature similarity in the spatio-temporal dimension. Using a graph community detection algorithm, we find tightly connected token clusters in the graph, which represent stable and persistent visual elements in the video. We select a few central tokens with higher centrality from each cluster as representatives. The second part is **difference-based event token selection**: we pay special attention to the edges connecting along the time axis in the graph. When the similarity between corresponding tokens of adjacent frames drops sharply, we consider it a turning point and mark these tokens as critical event tokens that must be

retained. Finally, we merge these two sets of tokens to be input into the LLM. Our contributions can be summarized as follows:

- We are the first to propose that the similarity and difference of tokens should be given equal importance in VLM efficiency research.
- We design a spatio-temporal graph framework to uniformly model the complex spatio-temporal relationships between video tokens. We propose a novel dual token selection strategy that can simultaneously screen for both representative and pivotal event tokens.
- We conduct extensive experiments across three long-video benchmarks and two base models. ST-SimDiff ranks first in nearly every comparison: at a 30% token budget it already matches the full-token upper bound on VideoMME (63.2% vs. 63.3%), and at a 50% budget it surpasses it (63.8%), all while cutting inference time and peak memory by up to roughly 30% relative to the uncompressed model.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has significantly propelled progress in multimodal understanding, leading to the emergence of numerous prominent Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These LVLMs typically process images or video frames by converting them into visual tokens via pre-trained visual encoders, which are then fed into LLMs. Various alignment modules commonly connect the visual encoder and the LLM, including MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023). This architecture enables LVLMs to exhibit exceptional capabilities in multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a). Particularly in video understanding, they show promise in processing longer, higher-resolution complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

However, this powerful capability also introduces significant computational challenges, especially in video understanding scenarios. Due to the need to process continuous sequences of frames, the number of visual tokens grows exponentially, far exceeding that of static images, posing a severe challenge to the training and inference efficiency of LVLMs. Therefore, how to efficiently process massive video visual tokens and reduce computational overhead remains a critical issue in current LVLM research.

2.2 VISUAL TOKEN COMPRESSION

Driven by the substantial overhead of video processing, visual token compression has become a critical method for improving the efficiency of Large Visual Language Models (LVLMs). Existing methods primarily approach compression from two aspects, token importance and similarity, which correspond respectively to the importance-based and hybrid baseline groups we compare against in our experiments.

Earlier methods primarily focused on importance-based token pruning, aiming to identify and remove visual tokens that contribute less to model performance. These methods typically utilize attention scores or other feature metrics to quantify token importance, such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Subsequent works (Shang et al., 2024; Ye et al., 2025) have started to further consider token similarity to more effectively reduce redundancy and preserve critical information, recognizing that even important tokens can exhibit high similarity and thus informational redundancy. For instance, FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning. VisionZip (Yang et al., 2025) reduces redundancy by selecting dominant tokens and merging contextual information. VisPruner (Zhang et al., 2025b) leverages visual cues to remove redundant items. In our benchmarks these hybrid methods do prove stronger than importance-only pruning, with FrameFusion the most competitive prior approach overall; ST-SimDiff nonetheless surpasses all of them on both base models and at both token budgets.

Despite significant progress in visual token redundancy, existing work hardly addresses the intricate, global spatio-temporal similarity relationships between tokens. In video scenarios, visual informa-

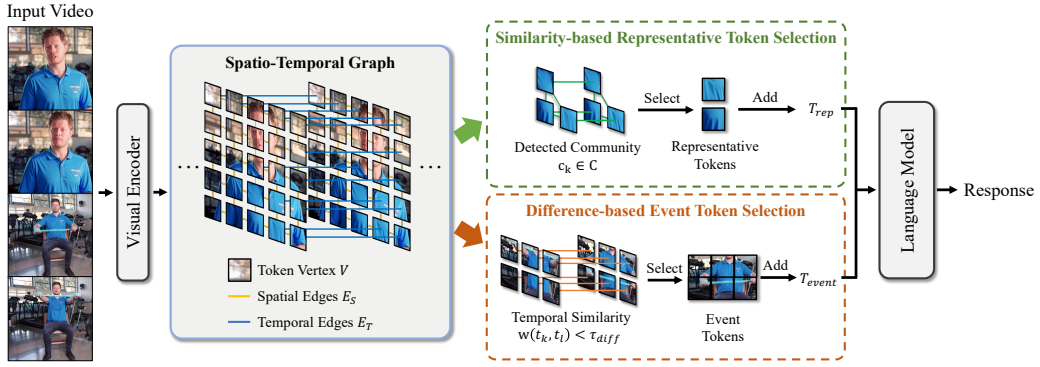


Figure 2: The overview framework of ST-SimDiff, which consists of three parts: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection.

tion is redundant both spatially and temporally. Existing similarity-driven methods, like Frame-Fusion, rely mainly on the temporal similarity between adjacent frames, failing to utilize complex spatio-temporal relevant information. This limits their ability to fully exploit inherent video redundancy, and the gap is most visible precisely in extreme token reduction scenarios: at the aggressive 30% budget where such methods have the least room to spare, ST-SimDiff opens its largest margins over them.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) first encodes the video into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This process is computationally demanding, as the self-attention mechanism’s complexity scales quadratically with the token count N , i.e., $O(N^2)$. Our task is therefore to design an efficient token selection function, $f(\cdot)$, that takes the full sequence T_{full} and a compression ratio r to produce a much smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The objective is to maximize the LVLM’s downstream task performance using this subset, subject to the constraint that its size $|T_{\text{sub}}|$ is approximately $r \cdot N$. The core challenge is to design the selection function $f(\cdot)$ to retain the most critical information within the subset T_{sub} .

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, the proposed ST-SimDiff method comprises two key components that operate in parallel on a spatio-temporal graph constructed from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). First, in the SRTS module, we identify and condense the video’s stable and redundant content. By applying community detection algorithms to the graph, we group highly similar tokens into clusters, which correspond to persistent visual elements like static backgrounds. From each cluster, we then select a few highly central tokens to form a representative set, T_{rep} . Second, the DETS module is designed to capture the video’s crucial dynamic shifts. It analyzes the temporal edges of the graph, identifying moments where the similarity between corresponding tokens in adjacent frames drops sharply. These points of high difference signify turning points, and the tokens framing these transitions are preserved as critical event tokens, forming the set T_{trans} . By synergistically combining these two components, ST-SimDiff generates a final token subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$, which is both highly compact and information-rich, retaining stable content while precisely capturing key events.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a sequence of tokens $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ corresponds to a specific spatio-temporal location and is associated with a feature vector $\mathbf{x}_k$. We define the location of each token t_k by its frame index $T(t_k)$, its spatial height index $H(t_k)$, and its spatial width index $W(t_k)$.

We model the relationships between these tokens using a spatio-temporal graph $G = (V, E)$, where the vertex set V represents all tokens in T . The edge set E is the union of two distinct subsets, $E = E_S \cup E_T$, which capture spatial and temporal relationships respectively. The spatial edges E_S connects tokens that are spatially adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connects tokens that are at the same spatial position but in adjacent frames, capturing temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

The weight of any edge $w(v_i, v_j) \in E$ is defined by the cosine similarity of the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph structure efficiently encodes local spatial relationships and temporal continuity, providing a unified foundation for our dual-path token selection strategy.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content is inherently characterized by substantial spatio-temporal redundancy. For instance, in a static scene that persists for several seconds, the background tokens exhibit high similarity over time. Similarly, for an object moving across the screen, its corresponding tokens, despite changing spatial positions, maintain a high degree of semantic correlation. Existing methods often address temporal and spatial similarities in isolation. In contrast, our spatio-temporal graph is designed to capture these joint similarities. Within this graph, semantically related tokens form dense ‘‘communities’’ or ‘‘clusters’’ based on their feature similarity, irrespective of their specific frame or location. Therefore, selecting representatives from each community emerges as a highly efficient strategy for compressing redundant information while preserving core semantics.

To accurately identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only the edges with weights above this threshold, forming a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to identify token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, we rank and filter its internal tokens based on their centrality. The centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is defined as its average similarity with all other tokens within the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

Subsequently, we set the intra-community retention rate to the globally preset compression ratio r , and preserve the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores from each community c_k . This uniform filtering process naturally handles all communities, including single-node communities composed of unique tokens (in which case the sole node is retained). Formally, the set of representative tokens, T_{rep} , is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. In summary, this strategy efficiently compresses the static and persistent content of the video by identifying and retaining the most central tokens within each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity defines the “norm” of a video, then difference defines its “events”. A model focusing only on similarity may excel at understanding “what is” but struggle with “what happened”. A video’s plot is driven by key events, and the essence of an event is change—the appearance of a new object, the start of an action, or a scene transition. In our model, these changes manifest as a sharp difference in the features of temporally adjacent tokens. Therefore, accurately capturing these “turning points” of significant difference is crucial for understanding the video’s dynamic content and correctly answering questions like “when” and “why”.

Difference, particularly along the temporal dimension, often signals the occurrence of a key event. For example, the entry of a new object, a scene change, or the beginning/end of an action will cause drastic changes in the visual features at corresponding positions in adjacent frames. We specifically analyze the temporal edges (E_T) in our spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ in the graph, which connects two temporally adjacent tokens t_k and t_l , we set a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores). When the difference score of a temporal edge (v_k, v_l) exceeds this threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, we consider the subsequent token t_l as a critical event token and retain it. Formally, the set of event tokens, T_{event} , is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

With this strategy, tokens that signify moments of significant temporal change can be preserved. By capturing these key transitions, we ensure that the crucial dynamic aspects of the video’s narrative are not overlooked during the compression process.

4.5 OVERALL REDUCTION PROCESS

In the proposed framework, we first compute the representative token set T_{rep} and the event token set T_{event} in parallel, and take their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, as the initial candidate set. To precisely meet the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$, we introduce a final pruning step. If the size of the candidate set, $|T_{\text{candidate}}|$, exceeds N_{target} , we remove the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance. Following existing work (Chen et al., 2024), the importance of a token is determined by its attention score in the shallow layers of the LLM. This step ensures that our method can flexibly meet any computational budget while preserving critical information.

In summary, our overall method adeptly combines two strategies: graph-based token selection and attention-based dynamic pruning. The former leverages the intrinsic structure of the video content (similarity and difference) to ensure the retention of core information, while the latter provides a flexible mechanism to precisely meet any given computational budget. This design makes our token compression process both principled and adaptive, thereby achieving a robust balance between efficiency and performance.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of our proposed ST-SimDiff framework can be analyzed in three main stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

The initial stage involves building the graph. For each of the N visual tokens, we compute the cosine similarity with its spatially and temporally adjacent neighbors. As each token has a small, constant number of neighbors, this step requires a single pass over all tokens, resulting in a complexity of $O(Nd)$, where d is the feature dimension of each token.

Following graph construction, the SRTS module identifies token clusters. As detailed in our implementation (Section 5.2), we employ a connected components algorithm, which operates with a near-linear time complexity of $O(N + |E|)$. Since the number of edges $|E|$ is at most $3N$, this simplifies to $O(N)$. The subsequent filtering step within each community could be computationally intensive. A naive approach of calculating all-pairs similarity within each community c_k would have a complexity of $O(|C| \cdot |c_k|^2 \cdot d)$. To prevent this, we impose a constraint on the maximum size of any community. In the rare event that a community exceeds a predefined threshold (e.g.,

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison on long-form video understanding benchmarks with LLaVA-Video-7B at 64 input frames (%). Rows are grouped by token retain ratio ($r = 30\%$ and $r = 50\%$) and compared against the full-token upper bound; the best score in each column within a group is in **bold**. ST-SimDiff attains the best overall accuracy in every group: 63.2% on VideoMME at $r = 30\%$ (equal to the 63.3% full-token upper bound while using $< 1/3$ of the tokens) and 63.8% at $r = 50\%$ (above the upper bound), with corresponding LongVideoBench-Overall scores of 57.5% and 57.9%. Its 30% margin over the strongest baseline (FrameFusion) is +1.9 on VideoMME-Overall, +1.5 on LongVideoBench-Overall, and +3.0 on EgoSchema.

$|c_k| > \sqrt{N}$), we partition it. This ensures the filtering process remains efficient and does not dominate the overall complexity. While other mainstream graph clustering algorithms such as the Louvain and Leiden methods offer more complex community definitions, their computational complexity is higher, approaching $O(N \log N)$. Therefore, we opted for the simpler and highly efficient connected components algorithm to ensure minimal processing overhead.

The DETS module involves a single pass through all temporal edges in the graph to identify significant changes by comparing token similarities against a threshold. This process has a linear complexity of $O(Nd)$.

In summary, the total complexity of our ST-SimDiff framework is governed by these linear-time operations, culminating in an overall complexity of $O(Nd)$. This scales far more favorably than the quadratic $O(N^2d)$ self-attention it replaces work for: the selection overhead grows linearly in N while the cost it removes grows quadratically, so the relative savings widen as videos get longer. This asymptotic advantage is borne out empirically: the time-saving rate over the uncompressed model rises from 23.0% at 32 frames to 30.2% at 128 frames, confirming that the framework delivers substantial token reduction at a computational footprint that is negligible relative to the model’s main inference cost.

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison on long-form video understanding benchmarks with NVILA-Video-8B at 64 input frames (%), reported under the same protocol as Table 1; the best score in each column within a token-budget group is in **bold**. The ranking transfers cleanly to this second base model: ST-SimDiff leads on VideoMME-Overall and LongVideoBench-Overall at both budgets, reaching 60.2%/55.2% at $r = 30\%$ and 61.7%/56.5% at $r = 50\%$. At $r = 50\%$ its 61.7% VideoMME-Overall exceeds the 61.5% full-token upper bound and improves on the best baseline by +0.9, while its 56.5% LongVideoBench-Overall likewise edges past the 56.3% upper bound and improves on the best baseline by +1.7.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To comprehensively evaluate our method’s effectiveness, we use LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as our base models and compare our approach against a range of state-of-the-art video token compression techniques under identical frame counts and token budgets, so that every reported difference reflects the compression strategy alone. These baselines cover both dominant strategy families: **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). To bracket the achievable range, we further include the original uncompressed model (Vanilla) as a performance upper bound and Random Sampling of an equivalent number of tokens as a lower bound, so that each method’s score can be read as a fraction of the accuracy recoverable at a given budget.

Benchmarks To test our method’s capabilities in long-video understanding, we adopt three challenging and complementary benchmark datasets that together stress different facets of the task: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). Reporting all three lets us show that ST-SimDiff’s lead is not benchmark-specific but holds across general video QA, long-range referring reasoning, and first-person causal understanding alike. VideoMME is a comprehensive benchmark for foundational video understanding, featuring diverse video types and three length categories: Short, Medium, and Long. LongVideoBench

evaluates the ability to retrieve and reason about details in long videos through a “referring reasoning” task, with videos of four lengths: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering dataset focused on long-term, first-person videos that tests the model’s ability to understand character intentions and causal chains of actions.

5.2 IMPLEMENTATION DETAILS

We use NVIDIA L20 GPUs with 48GB VRAM on Ubuntu 22.04, and run all inference evaluation through the *lmms-eval* (Zhang et al., 2024a) library so that every method is scored identically. For LLaVA-Video we follow its original 64-input-frame setting, and we fix NVILA to the same 64 frames so that the two base models are compared under matched computational and resource conditions. The two hyperparameters are held constant across all reported runs — similarity threshold $\tau_{\text{sim}} = 0.8$ and difference threshold $\tau_{\text{diff}} = 0.2$ (values justified by the robustness sweeps in the appendix) — while the token compression rate is varied over the two operating points $r = 30\%$ and $r = 50\%$ that define our comparison. For community detection we adopt connected-component finding, which our complexity analysis shows keeps the selection overhead linear while sacrificing negligible accuracy.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

To comprehensively evaluate our proposed ST-SimDiff framework, we compare it against a range of mainstream efficient video language model (VLM) compression methods. All experiments are conducted on three widely-used long-form video understanding benchmarks: VideoMME, LongVideoBench, and EgoSchema. To validate the generalization ability and robustness of our method, we report detailed performance data on two different base models, LLaVA-Video-7B and NVILA-8B, under token retain ratios of 30% and 50%. The comparison therefore spans $2 \times 2 = 4$ base-model $\times$ budget settings, and the results are presented in Table 1 and Table 2.

(1) **ST-SimDiff ranks first on overall accuracy in every one of the four settings, and does so by consistent rather than marginal amounts.** On LLaVA-Video-7B at $r = 30\%$, it reaches 63.2% on VideoMME-Overall and 57.5% on LongVideoBench-Overall, ahead of the strongest baseline (FrameFusion) by +1.9 and +1.5 points, and it leads on EgoSchema by +3.0 (56.0% vs. 53.0%). The same pattern holds at $r = 50\%$ (63.8% / 57.9% / 57.3%) and on NVILA-Video-8B (Table 2). Most strikingly, at the 50% budget the overall score not only beats every compression baseline but reaches the accuracy of the original model that keeps 100% of the tokens: 63.8% vs. the 63.3% upper bound on VideoMME for LLaVA-Video-7B, and 61.7% vs. 61.5% for NVILA-Video-8B. In other words, ST-SimDiff recovers essentially all of the full-token accuracy while discarding half the tokens, and even at the far more aggressive 30% budget it stays within 0.1 point of that upper bound on VideoMME.

(2) These margins become interpretable when read against the two baseline families. Among importance-based methods (e.g., FastV, FasterVLM), FasterVLM is typically the stronger of the pair, but both inefficiently handle the temporal redundancy prevalent in videos, retaining important yet repetitive tokens; this ceiling shows up as their consistently lower overall scores across both tables. Hybrid methods (e.g., PruMerge, VisionZip, FrameFusion) close part of that gap and are the more competitive group, with FrameFusion the strongest prior method overall. Yet even FrameFusion still optimizes for representative “commonality” information and carries a structural blind spot toward “turning points” and “abrupt events” — precisely where ST-SimDiff’s difference module adds its measurable edge, and precisely why our largest margins appear at the tight 30% budget where every retained token counts.

(3) Taken together, the numbers trace a clear progression: existing baselines reflect a deepening understanding of “redundancy,” yet remain limited on two fronts, the lack of synergistic spatio-temporal joint similarity analysis and the neglect of “difference” detection. ST-SimDiff addresses both — through a spatio-temporal graph that uniformly models complex spatio-temporal relationships and an explicit difference-detection mechanism — and the consistent, base-model-independent lead in Tables 1 and 2 is the direct empirical signature of closing these two gaps.

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component ablation on LLaVA-Video-7B (%), built up cumulatively from an importance-only Baseline by adding the similarity module (+Sim, broken into spatial-only, temporal-only, and joint variants) and then the difference module (++Diff, the full ST-SimDiff); the best row per column is in **bold**. Each addition helps monotonically: at $r = 30\%$, VideoMME rises 60.3%→62.6% (joint +Sim)→63.2% (++Diff), so the joint similarity design contributes +2.3 and the difference module a further +0.6 over the importance-only baseline. The difference module’s contribution is largest at the tighter 30% budget, consistent with its role of rescuing event tokens when the budget is scarce.

5.4 ABLATION STUDY

To validate the effectiveness of each core component within our ST-SimDiff framework, we conducted a series of detailed ablation studies. Starting from a Baseline that uses only fundamental importance-based pruning, we add the similarity selection module (+Sim) and then the difference selection module (++Diff), so that the incremental gain of each component can be read directly off successive rows. All experiments were conducted at token retain ratios of 30% and 50%, with the results in Table 5.

(1) The +Sim module produces a consistent, monotone gain over the baseline on every benchmark and at both budgets. Decomposing it into spatial-only, temporal-only, and spatio-temporal joint variants makes the ordering explicit: the joint variant is best in each case, and on VideoMME at $r = 30\%$ it reaches 62.6% versus 61.5% (spatial-only) and 61.7% (temporal-only) — a +2.3-point lift over the 60.3% baseline that neither single-axis variant attains. This quantifies why modeling spatio-temporal associations jointly, rather than one axis at a time, is the more effective way to compress redundancy.

(2) Adding the difference selection module (++Diff) on top of the joint similarity module completes the framework and yields a further gain that is sharply budget-dependent. At $r = 30\%$ it lifts VideoMME from 62.6% to 63.2% (+0.6), LongVideoBench from 57.0% to 57.5% (+0.5), and EgoSchema from 55.3% to 56.0% (+0.7); at $r = 50\%$ the same additions are smaller (e.g., 63.7%→63.8% on VideoMME). This contrast is itself informative: the difference module functions as a safety net that recovers unique event tokens the aggressive +Sim module discards, and its payoff is naturally largest under the tight 30% budget where such tokens are most at risk, whereas at 50% the more lenient +Sim already retains many of them and the scores are already near the upper bound. The pattern quantitatively confirms our core motivation — balancing similarity-based redundancy compression against difference-based event preservation matters most precisely in the demanding high-compression regime.

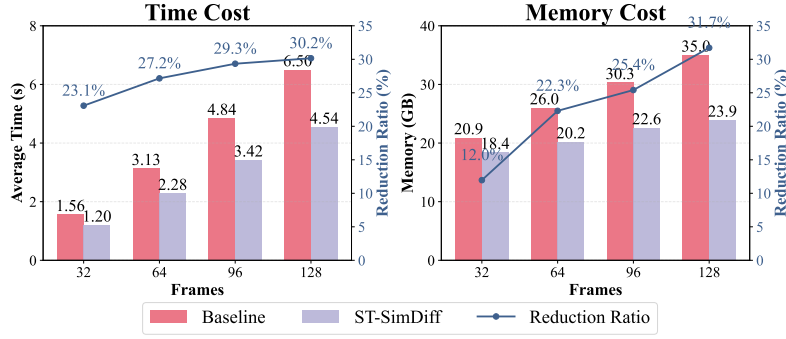


Figure 3: Computational cost of ST-SimDiff versus the uncompressed baseline LLaVA-Video model, measured as average inference time and peak GPU memory under a 30% token budget across 32–128 input frames. The two curves diverge as frames increase: the baseline’s inference time grows from 1.56s to 6.50s while ST-SimDiff holds it to 4.54s at 128 frames, and peak memory rises from 20.9 GB to 35.0 GB versus only 23.9 GB for ST-SimDiff. The resulting savings grow with video length — from 23.0% to 30.2% in time, and up to 31.7% in memory — showing the efficiency advantage compounds precisely where long-video processing is most expensive.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify the efficiency advantages of our proposed ST-SimDiff method in practical applications, we measured its average inference time and peak GPU memory usage under a 30% token budget, benchmarked directly against the same LLaVA-Video model run without any compression. Both metrics are reported over the same 32-to-128-frame sweep in Figure 3, so the two methods differ only in the token-reduction step.

Inference Time The speedup is not fixed but widens with video length. The uncompressed baseline’s average inference time grows sharply from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames, whereas ST-SimDiff curtails this to just 4.54 seconds at 128 frames. Expressed as a saving rate, this improves from 23.0% at 32 frames to 30.2% at 128 frames: the longer and more expensive the video, the larger the relative gain, exactly the regime where efficiency matters most.

Memory Cost The memory picture mirrors the timing one. The baseline’s peak GPU memory climbs from 20.9 GB at 32 frames to 35.0 GB at 128 frames, while ST-SimDiff caps it at 23.9 GB for the same 128-frame input — a 31.7% reduction — by cutting the token count fed into the LLM. This 11.1 GB absolute headroom is enough to push longer videos onto memory-limited hardware that the uncompressed model would not fit, broadening the range of deployable settings.

In summary, ST-SimDiff pairs its leading accuracy (Tables 1 and 2) with efficiency gains of up to 30.2% in time and 31.7% in memory, and both advantages scale up with video length — it is simultaneously the most accurate and among the most economical option in our comparison.

6 CONCLUSION

In this paper, we address the computational inefficiency of Multimodal Large Language Models when processing long videos, a problem rooted in the excessive number of visual tokens and the neglect of critical content changes. We propose ST-SimDiff, a framework requiring no training that builds a spatiotemporal graph to model complex associations between tokens, coupling a similarity-based selection that identifies redundancy via community detection with a temporal difference-based selection that preserves tokens at key transition points. The evidence for this design is consistent across the board: on VideoMME, LongVideoBench, and EgoSchema, and on both LLaVA-Video-7B and NVILA-Video-8B, ST-SimDiff ranks first in every token-budget setting we test, matching or exceeding the full-token upper bound (e.g., 63.8% vs. 63.3% on VideoMME at a 50% budget) while trailing it by at most 0.1 point even at the aggressive 30% budget, and it improves over the strongest prior method by margins as large as +1.9 points. These accuracy gains are accompanied

by up to 30.2% less inference time and 31.7% less peak memory, with both savings widening as videos lengthen. By quantifying that balancing content representativeness against transitional events is what drives these gains, this work opens a new and empirically grounded direction for efficient video representation.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

To make the mechanism behind these numbers concrete, Figure 4 visualizes the intermediate and final token selection on a sample sequence featuring a dynamic object-manipulation task. For the Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” show the graph community detection at work, grouping spatially and temporally redundant regions — the static background and stationary objects — into cohesive communities. SRTS then keeps only a few central “representative tokens” per community, the sparse yellow boxes in the final results, which is exactly the redundancy-compression step that our ablations credit with the joint +Sim gain.

Complementing this, the Difference-based Event Token Selection (DETS) activates during dynamic moments. In the “Final Results” row, tokens triggered by high temporal difference are marked with dense red boxes that track the moving hand and the shifting blue liquid, showing that the method locates the “turning points” where the temporal difference exceeds τ_{diff} . Preserving these tokens is what keeps fine-grained action detail intact, and it is the same event-recovery behavior responsible for the difference module’s largest measured gains under the tight 30% budget.

The final overlay makes the synergy visible: sparse yellow tokens carry the stable background while dense red tokens capture the rapid motion. This split is the qualitative image of the quantitative trade-off ST-SimDiff strikes — densely sampling during change to avoid the blind spot of similarity-only methods, while pruning the spatio-temporal redundancy that importance-based methods retain — and it explains, at the token level, why the framework recovers near-full-token accuracy at a fraction of the tokens.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Generalization to a third base model, Qwen2.5-VL-7B, at 64 and 128 input frames (%), reported against the full-token upper bound at both frame counts and both token budgets ($r = 30\%$ and $r = 50\%$). The efficiency-accuracy trade-off stays tight: at $r = 50\%$ with 128 frames, ST-SimDiff reaches 66.3 on VideoMME-Overall, within 0.1 of the 66.4 full-token upper bound, and 59.8 on LongVideoBench-Overall versus a 60.4 upper bound, while retaining only half the tokens.

To test whether the gains reported in the main paper are tied to a particular architecture, we repeat the evaluation on the recently released Qwen2.5-VL model, using the VideoMME and LongVideoBench benchmarks and the same two token budgets. This provides a third, independent base model on which to read off the efficiency-accuracy trade-off.

The results in Table 4 show the pattern carries over cleanly. Reading the 64-frame setting, at a 30% retention ratio ST-SimDiff-64 already reaches 62.4 on VideoMME and 56.8 on LongVideoBench, and at 50% it climbs to 62.9 and 58.4 respectively — the extra token budget buys a consistent +0.5 on VideoMME and +1.6 on LongVideoBench. Most telling is how close these compressed scores sit to the full-token ceiling: at $r = 50\%$ the 62.9 on VideoMME exactly matches the 62.9 upper bound of Qwen2.5-VL-64, and the 58.4 on LongVideoBench is within a point of its 59.2 full-token

counterpart. That a half-token budget reproduces near-full-token accuracy on a model the framework was never tuned for is direct evidence of architecture-independent generalization.

C ABLATION STUDY RESULTS ON NVILA

Method	VideoMME	LongVideoBench	EgoSchema	
Token Retain Ratio $r = 30\%$				
Baseline	59.3	53.8	50.2	
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)	60.2	55.2	51.7	
Token Retain Ratio $r = 50\%$				
Baseline	60.9	55.3	51.5	
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)	61.7	56.5	52.5	

Table 5: Component ablation on NVILA-Video-8B (%), following the same cumulative Baseline→+Sim→++Diff protocol as the main-paper ablation, with the best row per column in **bold**. The additive trend reproduces on this second base model: at $r = 30\%$, VideoMME rises 59.3%→59.9% (joint +Sim)→60.2% (++Diff), and the full ST-SimDiff is best in every column at both budgets, confirming that each module’s contribution is not model-specific.

To check that each component’s contribution generalizes beyond LLaVA-Video-7B, we repeat the cumulative ablation on NVILA-Video-8B (Liu et al., 2024c) under an identical setup, again starting from an importance-only Baseline and adding the similarity module (+ Sim) and then the difference module (++ Diff).

Table 5 shows the per-component trend is highly consistent with the LLaVA-Video-7B result, module for module. Adding + Sim yields a steady improvement at both the 30% and 50% budgets, and among its three variants the spatio-temporal joint strategy is the strongest: at $r = 50\%$ it reaches 61.4% on VideoMME, ahead of the spatial-only (61.1%) and temporal-only (61.3%) variants, quantifying the value of modeling spatio-temporal associations jointly rather than one axis at a time. Layering the difference module (++ Diff) on top completes ST-SimDiff and delivers the largest single jump in the table — to 60.2% / 55.2% / 51.7% at $r = 30\%$ and 61.7% / 56.5% / 52.5% at $r = 50\%$ across VideoMME, LongVideoBench, and EgoSchema. That both modules retain their measured contributions on a second base model is direct evidence that the gains stem from the framework’s design rather than any single model.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

The similarity threshold τ_{sim} and the difference threshold τ_{diff} govern representative- and transitional-token selection respectively, and Figure 6 quantifies how sensitive performance is to each. Both curves trace the same inverted-U shape — rising to a peak and then declining — but the fluctuation is small, indicating robustness across a broad range rather than a knife-edge optimum. The peak is well defined: on VideoMME, accuracy tops out at 63.2% at $\tau_{sim} = 0.8$ and again at $\tau_{diff} = 0.2$, with LongVideoBench peaking at 57.5% at the same settings. The tails are consistent with the mechanism — too small a τ_{sim} yields imprecise community detection while too large a value fractures semantic clusters, and for τ_{diff} too small a value admits noise while too large a value misses key events — yet across the entire tested range the swing stays within about a point,

so the choice is forgiving. We therefore fix $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, the joint optimum, for all other experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

To confirm that this hyperparameter robustness is not specific to LLaVA-Video, we repeat the threshold sweep on NVILA-Video-8B. Figure 6 charts the similarity threshold τ_{sim} and the difference threshold τ_{diff} against accuracy on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . Sweeping τ_{sim} over $[0.6, 0.95]$, accuracy moves within a narrow band and peaks at $\tau_{sim} = 0.8$ on both benchmarks — 60.2% on VideoMME and 55.2% on LongVideoBench. The drop-off away from the peak is slight, so the same $\tau_{sim} = 0.8$ that was optimal for LLaVA-Video remains optimal here, underscoring stability across a wide threshold range.

Impact of Difference Threshold τ_{diff} . Sweeping τ_{diff} over $[0.1, 0.4]$ yields the same picture: the optimum sits at $\tau_{diff} = 0.2$ on both benchmarks (60.2% on VideoMME, 55.2% on LongVideoBench), matching the value selected on LLaVA-Video. Setting the difference threshold near 0.2 is thus the most effective operating point for capturing key events under compression.

In short, both thresholds peak at the same values on both base models, and the surrounding fluctuation stays small. This cross-model agreement directly validates our main-experiment choice of $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$: the setting is not tuned per model but consistently optimal across benchmarks and architectures.

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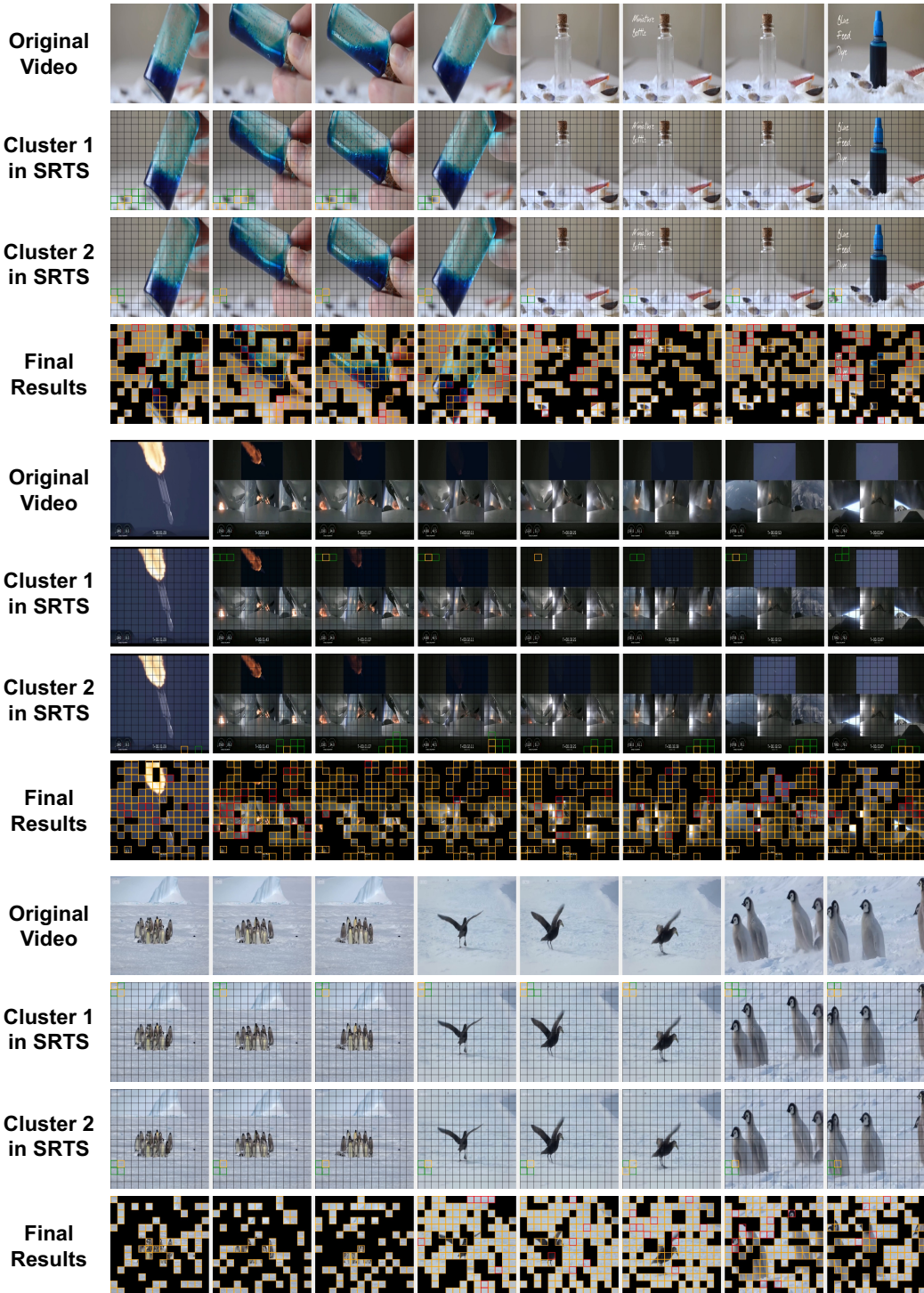


Figure 4: Qualitative breakdown of the ST-SimDiff token selection, complementing the quantitative results in the main paper. SRTS groups redundant regions into communities (grids in Cluster 1 and 2), while DETS flags event-driven tokens; the final overlay shows the resulting split between sparse representative tokens (yellow) for stable content and dense event tokens (red) for dynamic actions — a visual counterpart to the additive per-module gains measured in the ablations.

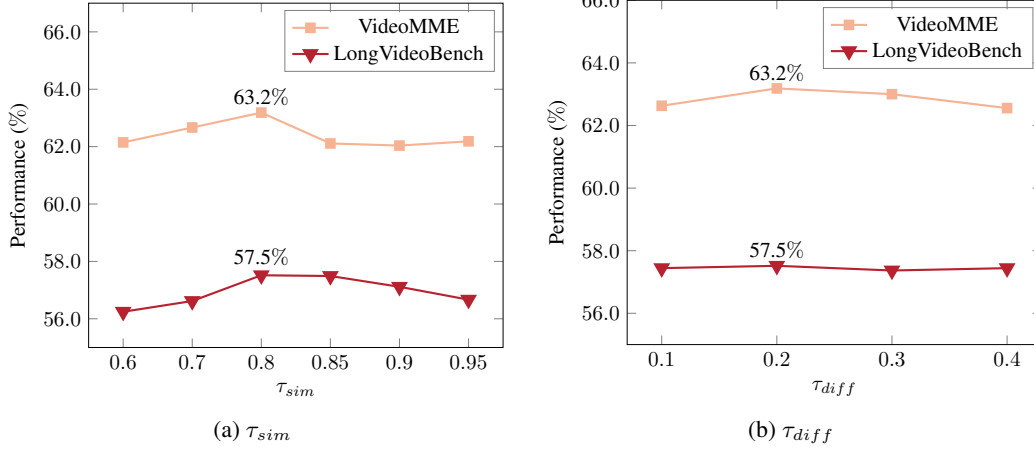


Figure 5: Ablation study results for different values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench.

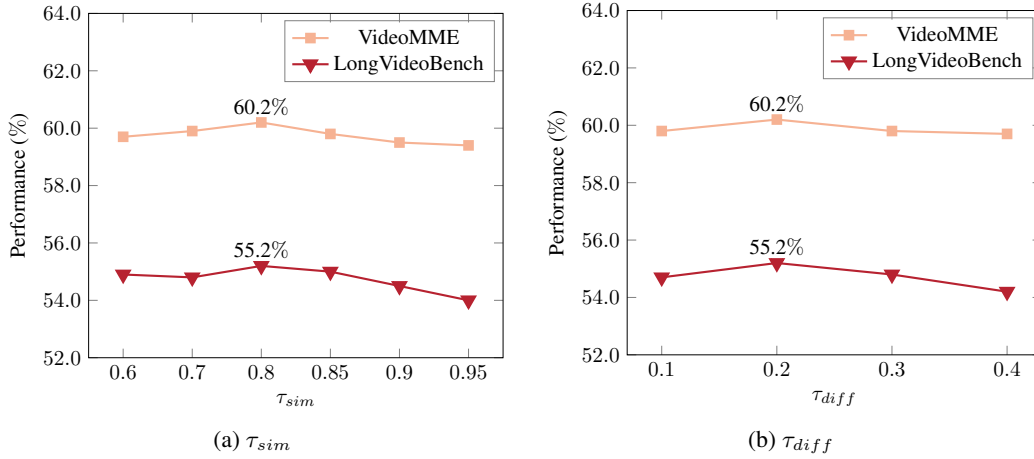


Figure 6: Ablation study results for different values of τ_{diff} and τ_{sim} on NVILA-Video-8B.